Enabling innovation in knowledge worker teams

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Abstract: Innovation and knowledge creation are processes, which require intensive collaboration of several people. Nevertheless, the constituents of successful collaboration in innovation activities are not well-known, and the understanding of the inherently social processes by which new intangibles are created is still in its infancy. This paper examines the key facilitators of innovation in knowledge-intensive group-level collaboration based on empirical evidence. The results demonstrate that social factors are highly influential for knowledge worker teams' ability to produce innovations. Teams whose internal processes were characterised by trust, vision, aim for excellence, participative safety and support for innovation were likely to attain high levels of innovation. Team diversity based on highly visible attributes was found to be negatively related with team innovation. Surprisingly, also external collaboration seemed to affect team innovation outcomes negatively, except for when it was aimed at general scanning for ideas and information. In addition, the differences between the more and the less innovative teams arose mainly from differences in their internal interaction processes, rather than patterns of collaboration with people outside the team.

Keywords: teams; groups; innovation; knowledge creation; knowledge workers; networks; collaboration.

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1 Introduction

While knowledge has always been at the heart of organisations, it is during the past two decades that the significance of knowledge as the most important source of economic value has really been unleashed (Foray, 2004). Knowledge has taken the place of land, labour and economic capital as the main source of corporate wealth creation and innovations have become the principal driver of competitiveness (Drucker, 1993; Castells, 1996; Quinn et al., 1997; Stewart, 1997). Knowledge workers are the fastest growing segment in the developed countries (Campion et al., 1996; Janz et al., 1997; Drucker, 1999, and they produce most of the value added in companies (Stewart, 1997). Drucker (1999) has even argued that increasing the productivity of knowledge workers will be the most important economic challenge of the 21st century.

Knowledge workers are highly educated employees who apply theoretical and analytical knowledge to developing new products, services, processes, and procedures (Drucker, 1999; Janz et al., 1997). The essence of knowledge work lies in the manipulation of symbols that convey data, information and ideas into new information and knowledge, in contrast to the manipulation of concrete materials (Reich, 1992). Consequently, knowledge workers must master, e.g., information processing, knowledge integration, knowledge creation and abstract thinking. Drucker (1997, p.23) proposes that knowledge differs from all other kinds of resources in that it constantly changes and develops, with the result that 'today's advanced knowledge is tomorrow's ignorance'. This is what makes knowledge workers and their ability to analyse, combine and create knowledge so important. Especially in highly turbulent environments, the capacity for producing continuous development through knowledge creation and innovation activities is an essential facet of knowledge work (Teece et al., 1997; Eisenhardt and Martin, 2000).

Knowledge work tends to be organised in teams and most of innovation and knowledge creation activities take place in collaboration among multiple actors (Nonaka and Takeuchi, 1995; Leonard and Sensiper, 1998). However, there are few studies that treat innovation as a group phenomenon (e.g., Guzzo and Shea, 1992; Burningham and West, 1995; Cady and Valentine, 1999). Furthermore, the factors that determine performance of knowledge worker teams are different from the preconditions of effectiveness in other types of teams (Janz et al., 1997; Drucker, 1999). Consequently, it is of utmost importance to generate knowledge on conditions which improve knowledge worker teams' capacity for innovation.

The paper at hand examines the antecedents of innovation in knowledge-intensive teamwork. As the subject of knowledge worker team innovation has received relatively little attention in the earlier literature, the theoretical framework is constructed with the intention of covering extensively the group-level factors influential to the phenomenon. Team innovation, team diversity, within-team interaction and external collaboration are all studied from several viewpoints. The posited hypotheses are then tested on a sample of 20 knowledge worker teams in a large Finnish-based telecommunications corporation, which produces services and enabling technologies. Finally, results are discussed and implications for enhancing knowledge-intensive teamwork are presented. The paper contributes to the literature on intellectual capital management and organisational learning by providing information on the contextual features which enable innovation and knowledge creation in knowledge worker teams.

2 Antecedents of innovation in knowledge worker teams

In everyday speech, innovation is often used to signify creativity. The concepts of creativity and innovation are, however, distinct from each other in that creativity is a cognitive process that occurs within the mind of a single individual, whereas innovation is a social process, in which the elements of the process are events that happen between people (West and Farr, 1990). Another important difference is that while creativity refers to generation of new ideas, innovation is a new idea put to use – which necessarily makes it a social phenomenon, because even though ideas can reside in the head of one individual, when they are acted upon, they become a part of social reality. The creativity of an individual may be seen as a starting point for innovation, although not all innovations are necessarily creative, nor does creativity always lead to innovation. Another concept that is closely related to innovation and is often confused with it, is invention. Invention, however, is a much narrower concept than innovation. Invention is a legal term that stands for a patentable improvement, which can be produced industrially, while innovation can be any novelty or reform, which can pertain not only to a product but also to for example processes, organisational structure, strategy or marketing.

Innovation has been used to denote the process through which something new is developed, as well as the end product of this process. Innovations as end products have been classified according to various attributes, such as their novelty (from incremental to radical), complexity, successfulness and the aspects of the organisational system that they impact (e.g., business concept innovations, managerial innovations) (West and Farr, 1990). Innovation has also been studied on different levels. King and Anderson (1990) distinguish between literature pertaining to individual, group and organisational innovation. Of the three levels of analysis, organisational innovation, and especially its antecedents, has received the most attention, whereas group level is by far the least studied.

West and Farr (1990) have defined innovation exhaustively as follows: 'Innovation is the

1 intentional

2 introduction and application

3 within a role, group, or organisation

4 of ideas, processes, products or procedures

5 new to the relevant unit of adoption

6 designed to significantly benefit

7 role performance, the group, the organisation, or the wider society.'

This definition includes many noteworthy points. First, innovation is restricted to intentional attempts to make changes. Second, the definition recognises that there are two components in innovation, namely the new idea and its implementation. Third, production of innovations can be examined in the level of individual, group or organisation. Fourth, innovations are not restricted to technical advancements or products, but can also be advancements in for example administrative practices. Fifth, it is not required that innovation be absolutely novel, simply that it is new to the relevant unit of adoption, as in cases where a new employee brings new ideas to an organisation from his or her previous job. Sixth, benefits derived from an innovation do not need to be financial, they can be, for example, improvements in the job satisfaction of employees. Finally, the definition allows for the introduction of a new idea designed to bring benefits not only to the individual, group or organisation, but also to society in general.

Next, we will overview factors which have been suggested to influence the capability of work groups to create innovations: team diversity, intrateam interaction and relationships with extra-team actors.

2.1 Team diversity, conflict and innovation

According to minority influence research tradition, intragroup conflict is a necessary precondition for innovation. In the classical colour perception study by Moscovici et al. (1969), innovation is actually defined as 'social pressure exerted by a minority'. According to this perspective, innovations are produced when a minority of a group's members succeed in inducing the majority to accept their views by acting in a coherent, persistent and confident manner (Moscovici et al., 1969; Mugny, 1982). Recently, it has been argued that minority opinions are beneficial for innovation not because of their better quality or higher truth-value compared with the majority opinion, but because of the thought processes that exposure to dissenting views stimulates. Exposure to differentiating views leads individuals to search for more information on the issue, to think more unconventionally and more divergently, i.e., to consider the issue from various perspectives. That is, minority dissent improves complex thinking, problem solving and creativity (Nemeth, 1997). In contrast, excessive conformity, characterised by Janis' (1982) concept of groupthink, undermines creativity and innovation.

Most of the writers on innovation, no matter what their school of thought, agree with the position that the conflict of diverse viewpoints is necessary for the emergence of new ideas (e.g., Kanter, 1988; West, 1990; Leonard-Barton, 1995; Nemeth, 1997; Amabile, 1998; Von Krogh, 1998). It is also widely agreed on that conflict has to be balanced with some kind of connectedness, lest other interpersonal processes important to innovation suffer (West, 1990; Nonaka and Takeuchi, 1995; Leonard and Sensiper, 1998; Brown and Duguid, 1998). The complementarity of cognitive conflict and positive team spirit is reflected in West's (1990) team climate factors of aim for excellence and participative safety. Nemeth (1997) talks about the fine balance of creating unity without uniformity. Also, Tjosvold's (1985) concept of constructive controversy and Leonard-Barton's (1995) creative abrasion comprise the same two-fold idea.

From the previous arguments, it can be gathered that to be conducive of innovation, intragroup conflict should be based on the diversity of intellectual opinions and not on personalised disagreements. Pelled (1996) suggests that diversity based on highly visible attributes such as race, age or sex is likely to lead to emotion-based disagreements, while diversity based on highly job-related attributes, such as education or organisational tenure lead to intellectual conflicts. There is some empirical evidence that diversity in highly job-related attributes improves groups' cognitive task performance and diversity in highly visible attributes impedes it (Pelled et al., 1999). Emotional and task-related conflicts were not directly addressed in the current research, but it will be assumed that the impacts of different kinds of team diversities on team innovation are congruent with Pelled's model.

However, in contrast with this track of thought, Milliken and Martins (1996), who conducted an extensive review of empirical studies on diversity published during 1989–1994, conclude that any kind of within-group diversity has negative affective consequences. This also applies to diversity based on attributes that are high in job-relatedness and low in visibility. In addition, they conclude that diversity in highly visible attributes has been associated with some positive effects on cognitive group performance. Thus, Pelled's model has not received unequivocal support.

As there are disparate claims and little empirical research (Cady and Valentine, 1999) concerning the impact of team diversity on innovation, their relationship shall be addressed in the present study. This will be done by looking at two highly visible forms of diversity, i.e., sex and age, and two highly job-related types of diversity, i.e., organisational tenure and educational field.

Hypothesis 1(a) Team diversity based on highly visible attributes diminishes team innovation.

Hypothesis 1(b) Team diversity based on highly job-related attributes improves team innovation.

2.2 Intrateam interaction and innovation

Team climate denotes either the cognitive schema or the shared perception of the working environment that is co-constructed in interaction among the team members while they are pursuing their common goals (Anderson and West, 1998). West (1990) has created a theory of team climate for innovation, which sophisticatedly combines findings of a wide array of earlier studies on group-level innovation and places them in a coherent frame of reference. According to him, the climate of innovative groups is characterised by four factors:

1 Vision

 'Idea of a valued outcome that represents a higher order goal and a motivating force at work'. Vision has four components: clarity, visionarity, attainability and sharedness. Clarity refers to the degree to which group goals are comprehended, visionarity to the extent that they are valued, attainability to the likelihood of their practical achievement and sharedness to the extent to which they are accepted by members of the group. Work groups with clearly defined and valued objectives are likely to develop new goal-appropriate methods of working because they are motivated and their efforts have focus and direction.

2 Aim for excellence

 General commitment to produce high-standard work that creates a group environment in which practices are questioned in a constructive manner in order to find possibilities for improvement, and differing viewpoints are cherished rather than suppressed.

3 Participative safety:

 Group members are encouraged to take part in decision-making and interpersonal relationships within the group are perceived as non-threatening. The more people participate in decision-making through interaction and information sharing, the more likely they are to commit to the outcomes of the decisions and to offer new ideas about improvements to be made. In addition, as expressing new ideas can be perceived as risk-taking, safe interpersonal climate facilitates innovation by minimising the risk of being ridiculed, punished or ignored.

4 Support for innovation

> 'The expectation, approval and practical support of attempts to introduce new and improved ways of doing things in the work environment'. For supporting innovation, it is not enough that the norms in favour of innovation are merely articulated, they also need to be enacted in form of active helping behaviour.

West's theoretical construct and the associated measure, Team Climate Inventory (TCI), have received considerable empirical support (e.g., Agrell and Gustafson, 1994; Burningham and West, 1995; West and Anderson, 1996; Kivimäki et al., 1997; Anderson and West, 1998; West and Smith, 1998). TCI has been found to reliably predict team innovation. In addition, Burningham and West (1995) found that the climate factors were related to team innovation more strongly than attributes of the individual group members, such as creativity, or structural characteristics of the teams, such as team size. However, the method has not been tested on knowledge worker teams specifically.

Hypothesis 2(a) Vision, aim for excellence, participative safety and support for innovation facilitate team innovation.

There is widespread agreement among organisational researchers that trust is critically important to social processes such as cooperation, coordination and knowledge sharing as well as for the general functioning of work-related relationships, be they on interpersonal, intra- or interorganisational level (e.g., Kramer and Tyler, 1996; Costigan et al., 1998). Furthermore, it has been claimed that interpersonal trust is a significant, if not even a necessary precondition of innovation (Nonaka and Takeuchi, 1995; Amabile, 1997; Jones and George, 1998; Nonaka and Konno, 1998; Von Krogh, 1998). However, no studies empirically addressing the relationship of trust and (group-level) innovation could be located in the literature review.

Hypothesis 2(b) Trust increases team innovation.

Cummings and Bromiley (1996) define trust as the belief that others

a make good-faith efforts to behave in accordance with their commitments

b are honest in negotiations

c do not take excessive advantage of others.

Mishra's (1996) definition of trust bears great similarities to the Cummings and Bromiley model: according to him, trust is willingness to be vulnerable to another party based on the belief that the other is

a reliable, i.e., there is consistency between actions and words

b open and honest

c concerned about the trusting subject's well-being

d competent.

Both definitions are multidimensional and encompass cognitive, effective and behavioural aspects of trust. The major difference between the two is that Mishra's construct has a fourth dimension lacking from the former conceptualisation, namely competence. As the nature of research targets' work is highly demanding and complex, i.e., characterised by uncertainty and low divisibility of activities (Kivimäki et al., 1997, p.376), it is likely that beliefs about other team members' competence influence the decision to trust them.

West's (1990) construct of team climate for innovation does not encompass the idea of interpersonal trust. At first glance, the dimension of participative safety might seem to be quite close to the idea of trust. However, when West's theoretical construct and items of TCI are inspected more thoroughly, it becomes evident that participative safety comprises acceptance, mutual influence and information sharing, but not trust *per se*. Interpersonal trust is something distinct from participative safety, even though surely related to it in many ways. Thus, it is argued here that interpersonal trust forms a dimension of team climate for innovation, which has been ignored so far in both theorising as well as empirical research on the subject.

2.3 External collaboration and innovation

It is widely agreed in the literature that collaboration across boundaries within the organisation increases innovation markedly. To name a few examples, Rosenfeld and Servo (1990) state that the main reason for why companies fail to implement potentially successful innovations is communication gaps between different parts of the organisation. According to Eisenhardt and Tabrizi (1995) close and frequent interactions among members of different organisational functions and units lead to innovation efficiency. Kanter (1988) argues that the most effective way to facilitate innovation is to create an integrative organisational structure based on collaboration, teamwork and multiple connections between parts of the organisation. Quinn et al. (1997) note that highly innovative organisations are characterised by high interconnectivity among individuals and permeable boundaries across work groups.

Furthermore, it seems that boundary spanning relationships within organisations are especially important for knowledge worker teams in rapidly changing business areas, such as the subjects of the current research. Campion et al. (1996) found that the amount of cooperation and communication between groups was more strongly connected to effectiveness measures for knowledge worker teams than for teams of other kinds of employees. Based on this, they arrived at the conclusion that knowledge workers have more need for integration with other parts of the company than other kinds of workers. Also, Tushman (1977) found that project teams with complex information processing requirements had more need for communication across group boundaries than teams facing less uncertainty and more stabile environments. Brown and Eisenhardt (1997) sum up the results of their case studies on several large-scale information technology companies by stating that those companies which can manage rapid, continuous change and innovation are characterised by extensive real-time communication among different parts of the organisation. Based on preceding arguments, Hypothesis 3(a) is as follows:

Hypothesis 3(a) Intra-unit and intra-organisational collaboration increase team innovation.

Ancona and Caldwell (1990, 1992) have studied activities and strategies that innovative teams adopt for handling external relations. They argue that team performance and innovation are not determined by the amount and frequency of external communication as such, but by its contents. Based on empirical results of several studies on research and development teams in high-technology organisations, they conclude that high-performing innovative teams assume the following two sets of activities toward others in the organisation outside their team:

1 Ambassadorial activities

  Protecting the team from outside pressure, persuading others to support the team, lobbying for resources, communicating with top management

2 Task-coordinator activities

  Coordinating and negotiating with external groups, obtaining feedback from others, communicating laterally with other functions

Ancona and Caldwell (1992) also state that engaging in

3 scouting activities, i.e., general scanning for ideas and information about competition, markets or technologies,

is detrimental to innovation. This is in contrast with the arguments brought up in the previous section that uncritically assume that a high quantity of interaction is beneficial for innovation, regardless of its purpose and content. Ancona and Caldwell offer three explanations for why scouting is likely to result in low team performance. Firstly, teams characterised by a high level of scouting direct their attention constantly to what happens in the environment, and this may make them unable to commit to any specific project and deadline. Secondly, concentration on scouting activities may reduce the effort that team members put into processes within the team and into external activities that are more directly relevant for high performance. Thirdly, scouting may indicate that the team's task is too general or too demanding, in which case, prolonged scouting demonstrates a futile search for an answer to an impossible question. The hypotheses concerning external activities are:

Hypothesis 3(b) Team innovation is facilitated by externally oriented ambassadorial and task-coordinator activities.

Hypothesis 3(c) Team innovation is diminished by externally oriented scouting activities.

3 Methods

3.1 Participants and procedure

Several teams and business units in a large Finnish-based telecommunications corporation were contacted and invited to take part in the research. A total of 106 questionnaires were distributed, and 88 (83%) of them were returned. The age of respondents ranged from 21 to 53 years (M = 30, SD = 6.64), and 59 (67%) of them were male. Respondents belonged to 20 teams. The team size ranged from 3–14 members (M = 5.30, SD = 2.66). On the average, 86% of the members of a team participated in the study.

The teams that participated in the research worked in areas of research and development, product management, customer relations management, knowledge management and administration. All teams' tasks required a high level of formal knowledge and analytical skills, as well as the skill to apply these to changing internal and external customer needs and development of new products, services and procedures. Thus, they all fit the description of knowledge worker teams. Except for one managerial team, the teams were situated on the same level in the organisational hierarchy. Team leaders were included in their respective teams.

3.2 Independent measures

Numeric diversity measures. Within-team diversity in relation to age and organisational tenure was computed using the coefficient of variation (standard deviation divided by the mean).

Categorical diversity measures. Within-team diversity based on sex, educational field and educational degree was measured with an entropy-based formula by Teachman (1980).

TCI short version. The shortened 14-item version of the original 38-item TCI was used to measure vision, aim for excellence, participative safety and support for innovation (Kivimäki and Elovainio, 1999). All items are worded with the team, not the individual, as the referent. The response format is a 5-point Likert-type scale. The coefficient alpha, was .89 for the whole scale, and .74, .74, .80 and .73 for each of the subscales respectively.

Trust. To measure the climate of trust in teams, a scale of five questions was developed based on Cummings and Bromiley's (1996) and Mishra's (1996) definitions. All items were measured on 5-point Likert-type scale. Coefficient alpha for the whole scale was .76, demonstrating a sufficient internal reliability. As with TCI composites, the team scores for trust were computed by averaging the individual members' sum-scores for each team and dividing the score by number of items.

Amount of external collaboration relationships. Information about the amount of external collaboration relationships was acquired by asking the team members to list the initials of the people working

a in the same business unit (but not in the same team) with whom they collaborated at least once a week

b in the same organisation (but not in the same business unit) with whom they collaborated at least once a month.

The initials were counted and transformed to figures. Dispersion of the figures was quite large, as the amount of intra-unit relationships ranged from 0 to 50 (M = 8.23, SD = 8.65), and intra-organisational relationships from 0 to 150 (M = 11.66, SD = 21.83).

Individual relationship figures were not computed into team-level indices simply by averaging them. This would not have been suitable because arithmetic mean is vulnerable to extreme values. Instead logarithmic transformation was made to individual values to normalise the distributions, so that large outliers would not manipulate the results of subsequent analyses. Team means were computed from the log-transformed individual scores.

External activities. The measure for content of the external relationships was developed based on Ancona and Caldwell's (1990, 1992) 24-item questionnaire. Eight of the original items were used for the questionnaire, three of these described ambassadorial, three task-coordinator and two scouting activities. Included items were chosen on the basis of high factor loadings in Ancona and Caldwell's study, and their wording was slightly modified to suit the tasks and working environment of the participants in the current research. Respondents were asked to report to what extent they felt responsible for various activities with outgroup members. Items were rated on a 5-point scale.

Cronbach's alpha was .65 for ambassadorial activities and .84 for task-coordinator activities. The relatively low score of ambassadorial activities would have risen to .72 if the last item, 'convincing others that my team is doing important work', had been removed. However, the decision was made to maintain this item in the measure, for without it, the scale would not have exhaustively covered the theoretical definition of ambassadorial activities. Measure of scouting activities consisted of only two items, so calculating composite estimate for this scale was not relevant. Team level scores for the external activities were acquired by averaging individual members' sum-scores for each team and dividing the score by number of items.

Open-ended questions. Also, qualitative data was gathered to check on the ecological validity of the variables used to predict team innovation, as well as to see how exhaustively these variables depicted subjects' experience of enablers and barriers of innovation. The items were open-ended questions, such as: 'What things in your team facilitate developing and implementing new ideas?'

Table 1 Examples and classification of the responses to the open-ended questions

Classification category	*Current facilitators of innovativeness*	*Current barriers to innovativeness*	*Innovativeness could be improved by*
Vision	• Clear responsibilities • Realistic and coherent goals	• Continuous change and chaos in the organisation • Lack of direction	• More realistic goal-setting • More efficient planning and organising
Aim for excellence	• Drive for success, enthusiasm • Pioneering spirit	• Pressure to make financial profit regardless of quality	• Increasing open criticism and questioning
Participative safety	• Team spirit, relaxed atmosphere • Informal and open communication	• Lack of authority in things concerning one's own work • Personality conflicts	• Taking every one's views into account • Possibility to participate in decision-making
Support for innovation	• Good processes for dealing with innovations (patent engineers, etc.)	• Lack of time, too tight schedules • Under resourcing	• Systematically collecting ideas • Having support and follow-up for the implementation of ideas
Trust	• Mutual sense of responsibility, diligence	• Power struggles, mandate grabbing	• Ensuring every one takes care of their share of the work
External collaboration	• Cross-functional gatherings and networks • Knowledge sharing across borders	• Lack of communication and collaboration between teams, projects, units • Not Invented Here – phenomenon	• Increasing collaboration, minimising competition within the company

Responses were first tentatively classified to five categories. The categories employed were vision, aim for excellence, participative safety, support for innovation, and trust. After the first classification round, there were many statements left that did not fit into these categories. Most of them clearly dealt with collaboration with people external to the team. Thus, a sixth category was formed, which was called 'external collaboration'. Practically, all of the responses to the open-ended questions could be classified in the six categories. Only two comments did not match any of the categories, both of them dealing with inconveniences in the physical working conditions. Thus, it can be concluded that the questionnaire is ecologically valid and covers comprehensively the factors that the teams perceived to influence innovation. Examples of the comments and their classification are given in the Table 1.

3.3 Dependent measures

Self-reports of team innovation. Team members were asked to rate the innovativeness of their team on a 5-point scale. Innovativeness was defined as the ability to develop new applicable ideas that profit the company. Team level scores were calculated by averaging the individual evaluations.

External ratings of team innovation. As the main purpose of the current study was to identify factors facilitating innovation, it was of utmost importance that the assessment of team innovation would be as accurate as possible. To gain a more objective assessment of teams' innovative output than self-reports could provide, Team Innovation Evaluation Form was constructed. Following Agrell and Gustafson's (1994) model, semantic opposites linked to innovative, quantitative and qualitative aspects of team output were generated. Seven of the semantic opposites were related to team creativity (e.g., risk-taking – security-seeking), three to the quantity (e.g., efficient – inefficient), and three to the quality (e.g., high-quality – low-quality) of team production. The measurement was done on a 7-point semantic differential scale.

As noted earlier when discussing the concept of innovation, innovation consists not only of the ability to produce new ideas, but also of the capability to put the new ideas into practice, and to do this in a way that is useful or profitable. For this reason, it was important that in addition to items concerning the ability to generate new ideas, the measure of team innovation also included items that dealt with the quantity and quality of the team product.

For all teams except for one, the form was filled in by a person external to the team but knowledgeable enough about the team to be able to evaluate its performance. In most cases, evaluator was the team leader's supervisor, whose job description included reporting about the team to higher hierarchical levels. For one of the teams, the only available evaluator was the team leader.

Coefficient alphas were .88 for team creativity, .79 for quantity and .94 for quality scales. For the whole 13-item composite measure, labelled simply Team innovation, alpha was .93.

4 Results

The purpose of this research was to find out how team diversity, within-team interaction and external collaboration influence team innovation outcomes. Answers to the hypotheses were sought by using correlation analysis and analysis of variance. For analysis of variance, the teams were classified to three groups according to evaluations of their performance. Where the condition of homogeneity of variances was not fulfilled, the non-parametric Kruskall-Wallis' test was used instead of analysis of variance. The relative importance of internal versus external team processes was assessed with regression analysis.

Table 2 Pearson correlations among team innovation outcomes and other variables

Variable	*Self-rated team innovation*	*Team creativity**	*Quantity of team product**	*Quality of team product**	*Team innovation composite**
Age diversity	r = –.13 p = .30	r = –.25 p = .14	*r = –.45* *p = .02*	r = –.33 p = .08	r = –.34 p = .07
Sex diversity	*r = –.42* *p = .03*	r = –.25 p = .14	r = –.32 p = .08	r = –.31 p = .09	r = –.31 p = .10
Organisational tenure div.	r = –.09 p = .36	r = .03 p = .45	r = .19 p = .21	r = .09 p = .36	r = .09 p = .36
Educational field diversity	r = –.33 p = .08	r = –.20 p = .20	*r = –.44* *p = .03*	*r = –.51* *p = .01*	r = –.36 p = .06
Vision	r = .31 p = .09	*r = .48* *p = .02*	r = .27 p = .13	*r = .48* *p = .02*	r = .47 p = .02
Aim for excellence	*r = .41* *p = .04*	r = .21 p = .19	r = .16 p = .25	*r = .39* *p = .04*	r = .27 p = .13
Participative safety	*r = .78* *p = .00*	r = .35 p = .07	r = .21 p = .18	r = .33 p = .08	r = .34 p = .07
Support for innovation	*r = .75* *p = .00*	r = .27 p = .13	r = .17 p = .23	*r = .42* *p = .03*	r = .31 p = .09
Trust	*r = .60* *p = .00*	r = .35 p = .06	r = .21 p = .19	*r = .45* *p = .02*	*r = .38* *p = .05*
Intra-unit relationships	*r = –.38* *p = .05*	r = –.36 p = .06	r = –.32 p = .08	*r = –.41* *p = .04*	*r = –.40* *p = .04*
Intra-organisational relationships	r = .01 p = .48	r = –.16 p = .25	*r = –.40* *p = .04*	r = –.33 p = .08	r = –.27 p = .12
Ambassadorial activities	r = –.09 p = .36	r = –.01 p = .48	r = –.11 p = .32	r = –.16 p = .26	r = –.08 p = .38
Task-coordinator activities	r = .24 p = .16	r = .23 p = .17	r = .14 p = .28	r = .15 p = .26	r = .21 p = .19
Scouting activities	*r = .56* *p = .01*	*r = .54* *p = .01*	r = .15 p = .26	*r = .43* *p = .03*	*r = .47* *p = .02*

Notes: 1-tailed tests of significance.
*Based on external ratings.

Hypothesis 1(a) posited a negative relationship between team innovation and diversity based on differences in age and sex. Correlations between age diversity and quantity of team product as well as between sex diversity and self-rated team innovation were statistically significant at $p < .05$ ($r = –.45$ and $r = –.42$). For both age and sex diversity, all correlations with team product variables were negative. Mean differences in age diversity were significant among groups rated low, moderate or high on team creativity (F = 8.91, p = .002), quantity of output (F = 6.69, p = .007), and quality of output (chi-square = 7.21, p = .03). Significant mean differences in sex diversity were detected between the different groups of output quality (chi-square = 5.63, p = .06), team innovation (chi-square = 6.76, p = .03), and self-rated team innovation (chi-square = 7.19, p = .03). All these findings were in line with the Hypothesis 1(a).

Hypothesis 1(b) posited a positive relationship between team innovation and team diversity based on educational field and organisational tenure. Correlational analysis showed that educational field diversity was negatively related with all performance measures, although only relationships with quantity and quality of team product were significant (r = –.44, p = .03 and r = –.51, p = .01). Furthermore, Kruskall-Wallis test showed that educational field diversity was significantly larger in teams that produced lower quality output than in teams that produced high-quality output (chi-square = 6.26, p = .04). Also, teams high on team innovation were more homogenous in relation to their members' educational field than teams rated low on team innovation (chi-square = 5.96, p = .05). These results are in fact directly opposed to Hypothesis 1(b). Organisational tenure diversity had no significant relations to any of the performance measures. Thus, it can be concluded that Hypothesis 1(b) received no support.

Hypothesis 2(a) predicted positive relationship between the four team climate variables and innovation outcome measures. In line with the hypothesis, all of the correlations were positive. Relations of team climate variables and innovation outcome measures were further explored by conducting one-way analyses of variance. The pattern of mean differences was clear: in all cases, means of the low innovation outcome groups were lower than means of high innovation outcome groups. Differences in the group means reached a significant level at $p \leq .05$ in five of the 20 explored cases (five innovation outcome scales × four team climate variables), and approached significance at $p < .1$ in six cases. In summary, Hypothesis 2(a) was supported.

Hypothesis 2(b) posited a positive relationship between trust and innovation outcomes. All correlations between trust and innovation outcomes were positive, and most of them reached a significant level. Mean differences of innovation outcome groups displayed a similar pattern to that found for team climate variables: the mean of trust was higher for all high innovation outcome groups than for low innovation outcome groups. The mean difference was significant for team creativity (F = 4.11, p = .04), quantity of output (F = 4.1, p = .04) and self-rated team innovation (F = 5.26, p = .02). In addition, it approached significance for both quality of team product (F = 2.67, p = .1) and team innovation (F = 2.67, p = .1).

Hypothesis 3(a) postulated positive relations between intra-unit and intra-organisational collaboration and innovation outcomes. Counter to the hypothesis, all of the correlations were negative, except for the association of intra-organisational collaboration and self-rated team innovation which approached zero (r = .01). The negative relations were significant for intra-unit collaboration with quality of team product (r = –.40, p = .04), team innovation (r = –.40, p = .04) as well as with self-rated team innovation (r = –.38, p = .05). Intra-organisational collaboration was significantly related with quantity of team product (r = –.40, p = .04).

Hypothesis 3(b) posited that ambassadorial and task-coordinator activities are positively related to innovation outcomes. Ambassadorial activities were negatively and non-significantly correlated to all innovation outcomes. Task-coordinator activities had positive relation to all outcome measures, but none of the correlation coefficients approached statistical significance. Thus, the Hypothesis 3(b) was rejected.

Hypothesis 3(c) predicted that scouting activities are negatively related to team innovation. Based on the results, it seems that the relationship was in fact directly opposite to that hypothesised. Scouting activities had the biggest number of significant correlations (four out of five) with innovation outcomes of all the independent variables, and all of them were positive. One-way analysis of variance showed that teams that rated themselves as more highly innovative engaged more in scouting activities than teams that rated themselves as low or moderate on team creativity ($F = 17.9$, $p < .001$). The mean difference was nearly significant for team innovation ($F = 2.94$, $p = .08$). In addition, the mean of scouting activities was higher for groups rated high than for groups rated low also in all other innovation outcome scales. Thus, Hypothesis 3(c) was rejected.

In addition to the posited hypotheses, multiple regression analyses were conducted to examine the relative impact of within-team interaction and external collaboration on team innovation outcomes. As the sample size was small ($N = 20$), it was necessary to combine independent variables into composites to avoid the risk of an overfit model. The composite of within-team interaction was constructed of the team climate and trust scores, and the composite of external relationships included task-coordinator and scouting activities, as all of these were positively related with team outcomes.

To inspect the relative importance of within-team interaction and external processes, the two composites were regressed onto team innovation outcome variables. Assumptions for regression analyses were evaluated from graphs. The regression residuals were distributed normally, and no heteroscedasticity or outliers were detected. The relative importance of predictor variables was assessed on the basis of standardised regression coefficients. The regression results are presented in Table 3.

The full regression equations, which included both within-team interaction and external relationships, accounted for a significant proportion of all innovation outcome variables, except for the quantity of team product. As sample size influences to a great extent the probability of finding significant relationships in regression analysis, it was decided to use significance level of $p \leq .1$ as criterion for acceptable predictive power. The relationship was particularly strong for self-rated team innovation: the model accounted for 56% of its variation. Based on these results, it seems like team creativity, quality of team product and team innovation, be it externally or self-rated, can all be quite well predicted by scores on within-team interaction and external relationship scales.

Examination of the individual impacts of the predictor variables showed that external relationships did not contribute significantly to the prediction of any team innovation outcomes. Within-team interaction scores were substantially related with self-ratings of team innovation, and they were also connected with quality of team product to a significant degree. Out of the five independent variables examined, only externally rated team creativity was better predicted by external relationships than by within-team interaction, and even in this case the difference between standardised regression coefficients was marginal. As with small samples, only very strong relationships can be detected with any degree of certainty, it is safe to say that within-team interaction is very highly related with self-rated innovation and quality of team product. One can speculate that had the sample been larger, also other patterns of connections might have emerged.

Table 3 Multiple regressions of within-team interaction and external relationship composites on innovation outcome measures ($N = 20$)

Dependent variable	*Independent variable*	β	R^2
Self-rated team innovation			.53***
	Within-team interaction	.61***	
	External relationships	.24	
Team creativity (externally rated)			.25*
	Within-team interaction	.29	
	External relationships	.32	
Quantity of team product (externally rated)			.06
	Within-team interaction	.21	
	External relationships	.09	
Quality of team product (externally rated)			.26*
	Within-team interaction	.43*	
	External relationships	.18	
Team innovation composite (externally rated)			.23*
	Within-team interaction	.34	
	External relationships	.26	

Notes: *$p \leq .10$, **$p \leq .05$, ***$p \leq .01$

5 Discussion

This study examined antecedents of innovation in knowledge worker teams. Overall, the results demonstrated that social factors are highly influential for knowledge worker teams' ability to produce innovations. As this research was among the first to empirically examine the significance of trust to team innovation, the most important single result was that interpersonal trust was related with high levels of team innovation. Also team climate was connected with high team innovation. Team diversity based on highly visible attributes was negatively related with team innovation. Surprisingly, also external collaboration had negative associations with team innovation, except for when it was aimed at general scanning for ideas and information. In addition, internal team processes were found to be more significant for team innovation than external processes. Next, the results concerning all facilitators and inhibitors of team innovation will be gone through in more detail.

The impact of team diversity on innovation

Surprisingly, the results of the current research speak for the superiority of homogenous teams, composed of individuals of same sex and age group, as well as similar educational background. The finding that educational diversity impedes innovation is somewhat surprising in the light of all the theoretical claims made for the necessity of divergent viewpoints for innovation.

Inspection of the descriptive demographic data of the seven teams rated externally as the most innovative showed that they were all highly homogenous: all of them consisted of males under 31 years of age with a technical education. Based on this, one is led to suspect that one reason for the finding that educational homogeneity leads to high innovativeness might lie in the target organisation's basic values. As the organisation was a telecommunications company, and most of the teams worked in business units or subsidiaries that were involved in developing enabling technologies, it may be that only technical advancements were perceived as innovative work by both the external evaluators and the team members themselves.

The impact of within-team interaction on innovation

The results clearly showed that highly innovative teams were characterised by vision, aim for excellence, participative safety, support for innovation and interpersonal trust. The fact that these same dimensions were constantly brought up in answers to the open-ended questions yields further credence in the role of these factors as facilitators of innovation in knowledge worker teams.

Unlike all other team innovation outcome measures, quantity of team product had no significant connections with any of the team interaction variables. This leads one to speculate that quality and innovativeness are more important indicators of knowledge worker productivity than quantity. Drucker (1999) among others takes this stance by arguing that continuing innovation forms an integral part of knowledge work, and that quality, rather than quantity is the essence of the output of knowledge work.

The impact of external relationships on innovation

The finding that numerous external relationships were related to low levels of team innovation outcomes was quite unexpected, for it directly contradicts much of the earlier theory and research referred to earlier. It may be that there is an optimum number of external connections, which some of the teams in the sample grandly exceeded. Unfortunately, this interpretation is impossible to validate with the current sample because of its small size.

Another possible explanation might lie in the teams' identities. Campion et al. (1996) found that knowledge worker teams with higher single-team identity, i.e., teams that perceived themselves as 'a group of members all working together as a single team' were higher on many effectiveness measures than teams which perceived themselves as 'a collection of individual employees doing their own work'. It might be that those teams in the current sample whose members had many external collaboration partners had weaker team identities, which in turn diminished their performance. As multiple group memberships, weakening of team boundaries and pressure to form networks are increasingly pervasive features of working life, the connection between team identity and external relationships would form an acute and interesting topic for another research.

Ancona and Caldwell (1990,; 1992) have argued that it is not the amount of outgroup interaction *per se* that influences team performance, but rather the content of these interactions. However, the content of external relationships had an opposite relationship with team outcomes than what Ancona and Caldwell found in their studies. In the studied sample, ambassadorial and task-coordinator did not enhance team innovation. The significance of scouting activities, on the other hand, was marked. This finding is in line with e.g., Nonaka and Takeuchi (1995) and Von Krogh (1998) who emphasise the significance of information flow across borders for innovation. Also, for example, Drucker (1997) states that gathering and analysing outside information is highly important, because it is the only means by which organisations can stay forerunners in the sudden shifts and rapid changes that increasingly characterise the global economy.

The most likely reason why the role of scouting emerged to be different in this research than in Ancona and Caldwell's (1992) study is that they studied new-product teams that were on late phases of product development cycle. Therefore, Ancona and Caldwell interpreted general scanning for external information and ideas to signify prolonged scouting, which no longer is appropriate once the final design is nearly closed down. In the research at hand, on the contrary, no data was gathered on work phase the teams were in, and indeed this could have posed serious difficulties because the sample consisted of teams with such various tasks. Nevertheless, it can be speculated that most of the teams were not in any strictly defined final phase of the innovation process, and therefore, that scouting was not inappropriate for their task performance. On the whole, this example serves as a reminder of the process nature of innovation: the facilitators of innovation are likely to be contingent on the phase of the innovation process.

Internal versus external processes and innovation

When the relative importance of within-team interaction and external relationships was compared, within-team interaction was found to be a more significant predictor of all team innovation outcomes except for externally rated innovativeness. This implies that for facilitating team innovation, the look should be first directed to what happens inside the team and only then to the team's connections with external partners, whether in the organisation or outside of it. Especially the team's self-assessment of its own innovativeness was substantially predicted by the way in which the team members perceived its internal processes. These results should serve as a warning against an unquestioning acceptance of the fashionable claims that networking is the best way to increase innovation. On the contrast, it seems like improving networks should be of little use for facilitating team innovation, if it is not done in parallel with developing the team internally.

Managerial recommendations

Based on the results, some practical recommendations can be given about the ways to facilitate team innovation. Firstly, differences between the more and the less innovative teams arise mainly from differences in their internal interaction processes. Thus, the attempts to improve team innovation should take the development of teams' internal functioning as their starting point. Teams whose internal processes are characterised by trust, vision, aim for excellence, participative safety, and support for innovation are likely to attain high levels of innovation. Secondly, networking with external partners is likely to be beneficial to innovation only if it is incorporated with the actual task that the team is striving to perform and does not threaten the team's internal processes. In any case, it seems like improving networks should be of little use for facilitating team innovation, if it is not done in parallel with developing the team internally. Thirdly, more diversity does not necessarily mean more new ideas: sometimes teams composed of members of same sex, age group and similar educational background produce more innovative results than heterogeneous teams.

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